

Shitesh Keshav Khaw

Full-Stack AI Engineer | FastAPI | Next.js | RAG | LLM Systems
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LinkedIn — GitHub — Portfolio

PROFESSIONAL SUMMARY

Full-Stack AI Engineer & AI Systems Architect specializing in scalable SaaS platforms, robust LLM integrations, and intelligent automation pipelines. Adept at decoupling complex architectures, managing cloud infrastructure, and bridging the gap between AI capabilities and resilient backend engineering to deliver high-availability, production-ready solutions.

TECHNICAL EXPERTISE

Languages: JavaScript, TypeScript, Python, SQL

Frontend: React.js, Next.js, Tailwind CSS, Framer Motion, Zustand, React Query, SWR

Backend: Node.js, Express.js, FastAPI, REST APIs, JWT Authentication, OAuth2, NextAuth, Zod, Pydantic

Databases: PostgreSQL, MongoDB, MySQL, Prisma ORM, ChromaDB

AI Engineering: Retrieval-Augmented Generation (RAG), Vector Search, Cross-Encoder Reranking, Semantic Search, Prompt Engineering, OpenAI API, OpenRouter, Groq API, LLaMA 3, Generative AI Systems

Cloud & DevOps: AWS (EC2, RDS, S3), Docker, Docker Compose, Oracle Cloud, Vercel, CI/CD, GitHub Actions, Nginx

Architecture & Systems: Multi-Tenant SaaS, RBAC, Audit Logging, Event-Driven Architecture, Distributed Systems, API Design, System Design

Testing & Quality: Jest, Unit Testing, Integration Testing, API Testing

Developer Tools: Git, GitHub, Postman, Linux, Claude, VS Code, Cursor, Antigravity

PROFESSIONAL EXPERIENCE

Full Stack Engineer — Makhija Quantum AI
Mumbai, India

Mar 2025 – June 2026

- Architected and deployed a highly available AI automation infrastructure, decoupling backend services from workflow orchestration to ensure scalable fault tolerance for B2B client onboarding.
- Engineered a high-throughput backend service layer using Python, eliminating manual lead-routing bottlenecks by automating 100% of incoming multi-channel communication workflows.
- Provisioned PostgreSQL databases on AWS RDS with a multi-tenant schema architecture, prioritizing strict data isolation and scalable ACID compliance.
- Designed a robust CI/CD strategy utilizing Docker across AWS and Oracle Cloud, maintaining continuous delivery and minimizing production downtime.
- Built complex, event-driven n8n automation pipelines processing multi-step data transformations across WhatsApp, Instagram, and LinkedIn webhooks with zero-downtime event logging.

KEY ARCHITECTURAL PROJECTS

DocuMind AI — Enterprise AI Platform

Next.js | FastAPI | PostgreSQL | ChromaDB | OpenRouter

Live Platform — GitHub Repository

- Architected an enterprise-grade Document Intelligence SaaS using Next.js, FastAPI, PostgreSQL, ChromaDB, and OpenRouter for compliance and auditing workflows.
 - Engineered a hybrid 2-stage RAG pipeline combining dense vector retrieval with Cross-Encoder reranking, significantly improving retrieval precision and citation quality.
 - Designed a multi-layer contradiction detection engine integrating deterministic validation with semantic AI analysis to identify inconsistencies across large document repositories.
- Built secure multi-tenant RBAC architecture with workspace isolation, immutable audit logging, JWT authentication, and explainable Trust Score analytics.

TaskOS — AI-Powered Productivity Platform

Next.js | PostgreSQL | Prisma | AWS | Groq API

Live Platform — GitHub Repository

- Architected a domain-driven Next.js 15.5 application utilizing a Neon Serverless PostgreSQL database, engineering 12 heavily indexed relational models with Prisma ORM for type-safe data transactions.
- Integrated LLaMA 3 70B via the Groq API to power a real-time AI productivity coach, implementing specialized domain layers for algorithmic task decomposition and streak-risk analysis.
- Engineered a robust authentication pipeline combining Google OAuth 2.0 with custom JWT middleware guards, enforcing rate limiting and Zod-based strict request validation across 14 API route groups.
- Designed a decoupled micro-service architecture to handle complex asynchronous tasks, including Web Push (VAPID) notifications, AWS S3 presigned avatar uploads, and Razorpay payment webhook processing.
- Developed custom, dependency-free SVG analytics pipelines to visualize task completion matrices and habit consistency heatmaps in real-time.

WordSage — Enterprise Writing SaaS

Next.js | Express.js | PostgreSQL | Docker

Live Platform — GitHub Repository

- Architected an enterprise-grade AI SaaS platform leveraging Next.js and TypeScript, utilizing a schema-first approach to ensure rigorous type safety across the full stack.
- Designed scalable REST APIs to execute compute-heavy LLM workflows, optimizing asynchronous execution to maintain sub-second response times for text expansion and summarization features.
- Engineered a secure, multi-tenant architecture featuring strict role-based access control (RBAC), effectively isolating data across user workspaces.
- Integrated an end-to-end monetized infrastructure via Razorpay, implementing secure webhook verification and encrypted OAuth session management.

EDUCATION

Bachelor of Science in Computer Science

Oct 2022 – Aug 2025

MVLU, Mumbai

HSC

Jul 2020 – March 2022

BHAVANS COLLEGE, Mumbai

SSC

Jul 2020

GNMHS, Mumbai